A Novel Deep Learning Approach to the Statistical Downscaling of Temperatures for Monitoring Climate Change

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ABSTRACT

General Circulation Models (GCMs) allow for the simulation of several climate variables through the year 2100. GCM simulations, however, are too coarse to monitor climate change at a local scale in a local region. Hence, one needs to perform spatial downscaling for these simulations, where statistical downscaling is often used. Statistical downscaling is performed by utilizing the large-scale GCM outputs to forecast a local-scale field (e.g., temperature). In this paper, we develop a new deep learning approach, named AIG-TRANSFORMER, which employs a novel attention-based input grouping (AIG) neural network followed by a transformer, for the statistical downscaling of the weekly averages of maximum (Tmax) and minimum (Tmin) temperatures using GCM-simulated climatic fields (climate variables). We formulate the downscaling problem as a multivariate time series forecasting task, with multiple GCM-simulated climatic fields as input features. We employ an attention mechanism within the AIG network to give selective importance to the input features while reducing the size of the input fed to the transformer. To test AIG-TRANSFORMER, we perform the statistical downscaling over the Hackensack-Passaic Watershed, in northeast New Jersey. We compare our new deep learning approach against several existing machine learning methods including random forests, support vector regression and long short-term memory networks. Experimental results show that AIG-TRANSFORMER outperforms the existing methods for downscaling both the maximum and minimum temperatures, with a Nash–Sutcliffe Efficiency coefficient of 0.84 for Tmax, and 0.85 for Tmin. We further apply AIG-TRANSFORMER to produce long-term projections over the 20 years period from 2030 to 2049, and report the annual means for maximum and minimum temperatures.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Machine learning approaches; • **Applied computing** → Physical sciences and engineering; Earth and atmospheric sciences; Environmental sciences.

KEYWORDS

Statistical Downscaling, Climate Change, Deep Learning, Transformer

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1 INTRODUCTION

The continuous increase of greenhouse gas emissions and the subsequent changes in climate variables are severely impacting communities' infrastructure and ecosystems. Climate change is affecting the operation and durability of key infrastructures, including bridges, roads, and power supplies. The quality and quantity of supplied water from freshwater ecosystems, along with the inhabitants of such ecosystems, are also in jeopardy because of climate change-driven events. These events mainly include the increasing severity and frequency of droughts, and changes in runoff dynamics. Changes in temperatures, and subsequently precipitation, are one of the leading causes of droughts and fluctuations of runoffs. Contingency plans for addressing climate change are of utmost importance for the sustainability and durability of communities' ecosystems and infrastructures. As such, predicting the changes in temperature contributes to more efficient disaster risk management.

To study climate change and future climatic projections, one often employs General Circulation Models (GCMs). GCMs are used to simulate climate systems at a global scale. Outputs from GCMs include historical, present, and future (~100 years into the future) time series climate variables. However, these models typically have very large spatial resolutions (>100km), making them too coarse to resolve the changes at a local scale in a local region. Moreover, the output of such models includes a lot of biases that need to be corrected in order to have efficient local climatic analysis. As such, to analyze and project regional climate changes, one would need to perform spatial downscaling and bias correction of GCM outputs.

Spatial downscaling of GCM outputs is performed using either dynamic [1] or statistical methods [2]. Dynamic downscaling is computationally expensive and requires advanced expertise in climatology. Statistical downscaling (SD), however, is less computationally demanding and could efficiently downscale multiple GCM outputs under different scenarios. Statistical downscaling approaches assume that the statistical relationship between local observations (e.g., temperature over a local region) and the large-scale GCM outputs (e.g., simulated temperature) is stationary. Moreover, such approaches rely on the assumption that the local climate patterns are largely influenced by the large-scale patterns. By efficiently downscaling GCM simulations, one would be able to better monitor climate change using long-term downscaling results.

To perform statistical downscaling, one would need first to extract the historical local observations (e.g., temperature over the Hackensack-Passaic Watershed), and select the desired GCMs to use, and extract their outputs under a certain scenario. Then, one would apply statistical approaches to relate the large-scale simulations to the local observations. A popular statistical approach is based on regression methods, including random forests (RF) [3, 4], support vector regression (SVR) [5, 6], and deep learning models [7-9] such as long short-term memory (LSTM) networks [10-12], among others.

In this paper, we develop a new deep learning approach that employs a novel attention-based input grouping (AIG) neural network followed by a transformer [13-15] for the statistical downscaling of the weekly averages of maximum and minimum temperatures. We refer to our new deep learning approach as AIG-TRANSFORMER. We formulate the downscaling problem as a multivariate time series forecasting task, with multiple GCM-simulated climatic fields as input features. We apply our deep learning approach to the New Jersey portion of the Hackensack-Passaic Watershed. By downscaling the maximum and minimum temperatures using AIG-TRANSFORMER, we would be able to provide long-term projections of temperatures. These projections are of great importance for practitioners and decision makers to better analyze and monitor climatic changes. The main contributions of our work are summarized below.

- We develop a new deep learning approach, named AIG-TRANSFORMER, that employs a novel architecture built specifically for the downscaling problem.
- Our experimental results demonstrate that AIG-TRANSFORMER outperforms several traditional machine learning methods and other deep learning models such as LSTM for the downscaling of the weekly averages of maximum and minimum temperatures.
- We further apply AIG-TRANSFORMER to downscale the daily maximum and minimum temperatures with good results.
- We use GCM future simulations as input to our trained AIG-TRANSFORMER to produce projections of maximum and minimum temperatures over the future period from 2030 to 2049.

2 THE PROPOSED APPROACH

2.1 Problem Formulation

Statistical downscaling of temperature is the use of GCM-simulated climatic fields (climate variables), such as precipitation, wind, and humidity over a large area, to predict the temperature over a smaller (local) region. GCM data are provided as a gridded dataset, and one usually extracts data from multiple grid points, referred to as atmospheric domains (represented by the nine black circles in Figure 1), surrounding the local region of interest (represented by the area highlighted in yellow in Figure 1). Subsequently, each GCM-simulated climate variable (e.g., precipitation) will appear in the input of a prediction model multiple times, corresponding to the grid points used (e.g., precipitation_point_1, . . ., precipitation_point_9). We formulate the statistical downscaling of maximum and minimum temperatures as a multivariate time series forecasting problem, and we aim to perform the downscaling for maximum and minimum temperatures separately and independently. The input of the prediction model consists of sequences of multiple GCM-simulated climate variables that are large-scale, over a large area, climatic fields taken from different grid points, and the output of the prediction model is a local, over a small region, maximum or minimum temperature. More precisely, we regard a local temperature dataset as a sequence of observations (ground truth values), where each observation consists of a timestep t, and a local maximum or minimum temperature value $T_{local,t}$. In obtaining the predicted local temperature $\hat{T}_{local,t}$ at time t, we configure a prediction model M to use current and previous large-scale climate variables, down to a specified time lag. Each climate variable has nine feature values extracted from the nine grid points in Figure 1. Let n be the total number of features. During training, we use sequences of training features $F_{i,j}$, $1 \leq i \leq n$, $(t - lag) \leq j \leq t$, over an interval from time t down to time $(t - lag)$ together with the label $T_{local,t}$ to train the model M. During testing, the trained M takes as input sequences of testing features $A_{i,j}$, $1 \leq i \leq n$, $(t - lag) \leq j \leq t$, over an interval from time t down to time $(t - lag)$ and produces as output $\hat{T}_{local,t}$ which represents either the predicted local maximum temperature (Tmax) or predicted local minimum temperature (Tmin) at time t. The default value of lag is set to 4.

2.2 Attention-Based Input Grouping

The input to a downscaling prediction model consists of large-scale climate variables (precipitation, humidity, wind, etc.) taken from different grid points. In standard machine learning downscaling approaches, each of these climate variables, for each grid location, is regarded as a unique feature (precipitation_point_1, precipitation_point_2, humidity _point_1, etc.), leading to a large number of features fed to the model for each timestep (225 features in our case). Having multiple features of essentially the same type (same

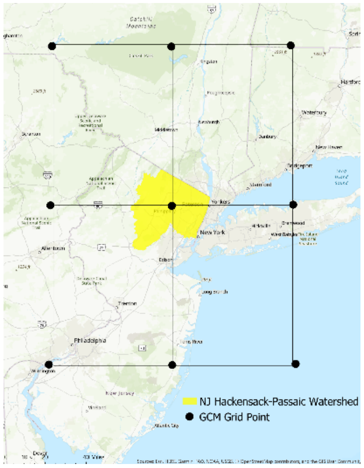

Figure 1: Atmospheric region of interest. The nine black circles represent the locations of the GCM grid points where large-scale GCM-simulated climatic fields (climate variables) are available and used as the input to a prediction model (e.g., AIG-TRANSFORMER). The yellow region represents the watershed area (local region) where a local maximum or minimum temperature will be predicted by AIG-TRANSFORMER.

climate variable) fed into a model might create bias and noise within the model, especially if the climate variable is highly correlated with the local temperature. We propose a novel approach, called attention-based input grouping (AIG), to encode similar features in one value that is then fed forward to the next layer of our deep learning model. In AIG, we cluster features (e.g., precipitation_point_1, . . ., precipitation_point_9) that belong to the same climate variable (e.g., precipitation) into a group, and feed the group to an AIG block. The aim of AIG is to group the same type of features together and perform selective attention on each feature group to produce one value per type/group.

As depicted in Figure 2, each AIG block consists of a 1D convolution layer, followed by a pooling layer, an attention layer, and feed-forward dense layers. The convolution layer aims to detect the local patterns of the corresponding feature group (e.g., precipitation), and the pooling layer extracts the patterns by focusing on the maximum, minimum, or average-based statistical summary of the neighborhood [16]. The main objective of the attention layer is to allow the AIG network to learn to choose a subset of the input features while producing the output. To achieve this, the layer saves the states from each feature in the input sequence and learn to give different weights to these features. As such, this layer reads the patterns produced by the pooling layer, and subsequently focuses on the relevant ones. The attention mechanism assigns a weight to the state of each feature, which depicts the state's relative importance. The weight α_{ij} between the target hidden state h_i and the source hidden state $\bar{h}_j$ is derived by comparing both states using the multiplicative form of a content-based function ($score(.)$) [17] and comparing the target state with each hidden state. The context vector c_i is the weighted sum of the source hidden states, and it captures the relevant input information. Finally, the attention vector v_i is the concatenation of the context vector with the last hidden state and activated using the hyperbolic tangent. The outputs produced by the attention layer are fed to fully connected neural networks, producing one single output value X_{G_i} for each feature group G_i.

2.3 AIG-TRANSFORMER

We adopt a variant of transformer for time series, in which we apply stacked multi-head attention blocks to the AIG output (see). Each multi-head attention block consists of a self-attention layer, a normalization layer, and a feed-forward layer. The prediction layer consists of three layers of fully connected neural networks and produces the predicted local temperature.

3 DATA

3.1 New Jersey Hackensack-Passaic Watershed

We choose the New Jersey (NJ) portion of the Hackensack-Passaic Watershed as the study area (the region highlighted by yellow in Figure 1). This watershed lies within 8 NJ counties, including Sussex, Passaic, Bergen, Morris, Essex, Hudson, Somerset, and Union. A relatively smaller portion of the watershed lies within the State of New York (not included in our study). The total area of the watershed is 590,546 acres, and its elevation ranges for up to 1,489 feet above sea level. The watershed houses around 650 farms, with most of them lying in Morris County. Facing global warming and the associated climatic changes, New Jersey in general is highly at risk with a predicted sea-level rise of 6.3 inches by 2100 [18].

3.2 Local Data

We retrieved the local maximum and minimum temperature data from the Global Historical Climatology Network (GHCN)-Daily. The GHCN dataset consists of data from thousands of stations worldwide and includes the data from the National Oceanic and Atmospheric Administration/National Climatic Data Center (NOAA/NCDC) [19]. Data consists of maximum temperature, minimum temperature, precipitation, wind, and other information, and can be requested from the NOAA NCDC online search tool. We extracted the maximum and minimum temperatures from January 1st, 1971, to December 31st, 2020 (total of 50 years). These observations (ground truth values) are from stations located in the watershed. The temperatures are averaged over all the stations, as we aim to perform downscaling over the entire watershed (NJ portion).

3.3 Large-Scale Data

We use the Meteorological Research Institute model from the Coupled Model Intercomparison Project Phase 5 (CMIP5) as our GCM model. We selected values of multiple climatic fields as potential features. These climatic fields (climate variables), described in Table 1, include temperature, wind, and humidity information, both near-surface and at different atmospheric pressure levels (200 and 500 hPa), in addition to cloud fraction, precipitation, sea level pressure, and heat flux information. We retrieved the GCM raw files for daily values, from year 1971 to 2020, from the data portals of the Earth System Grid Federation. Resolution of the GCM-used output is

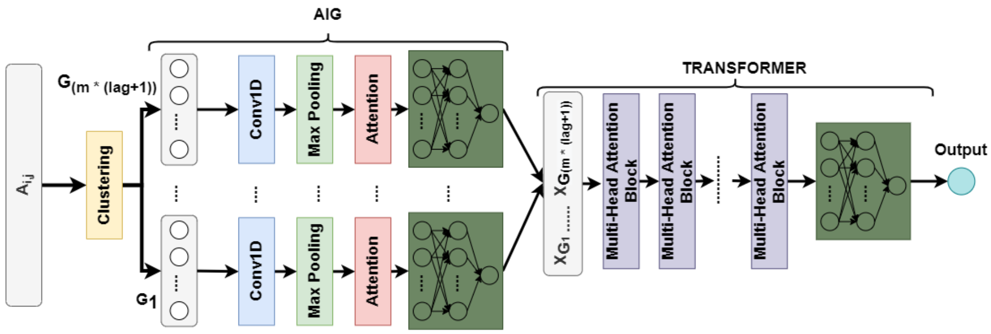

Figure 2: Architecture of the AIG-TRANSFORMER network. Time sequences of input features $A_{i,j}$, $1 \leq i \leq n$, $(t - lag) \leq j \leq t$, over an interval from time t down to time $(t - lag)$ are clustered for each timestep j into m feature groups based on their respective type where each group contains 9 feature values extracted from the nine grid points in Figure 1. There are $(lag + 1)$ timesteps, so there are $(m \times (lag + 1))$ groups in total. Each group is fed to an AIG block which consists of a 1D convolution layer, followed by a pooling layer, an attention layer, and fully connected dense layers. The outputs of all the $(m \times (lag + 1))$ AIG blocks are concatenated and fed into the transformer network. The final output value represents the predicted local temperature at time t. In this study, two AIG-TRANSFORMER networks are employed, one for predicting local maximum temperatures and the other for predicting local minimum temperatures.

Table 1: Large-scale climatic fields (climate variables) extracted from GCM model.

Climatic Fields	Elevation
Relative Humidity (%)	At surface, 250 hPa and 500 hPa
Specific Humidity	At surface, 250 hPa and 500 hPa
Precipitation (mm)	At surface
Wind Speed (m/s)	At surface
Air Temperature (K)	At surface, 250 hPa and 500 hPa
Eastward Wind (m/s)	At surface, 250 hPa and 500 hPa
Northward Wind (m/s)	At surface, 250 hPa and 500 hPa
Total Cloud Fraction (%)	In the sky
Surface Upward Latent Heat Flux (W m^{-2})	At surface
Surface Upward Sensible Heat Flux (W m^{-2})	At surface
Daily Maximum Relative Humidity (%)	At surface
Daily Minimum Relative Humidity (%)	At surface
Sea Level Pressure (Pa)	At surface
Daily Maximum Near-Surface Air Temperature (K)	At surface
Daily Minimum Near-Surface Air Temperature (K)	At surface

1.125^{o} x 1.112^{o}. We extracted the climatic fields (climate variables) presented in Table 1 for each grid point (represented by a black circle in Figure 1) contained in the atmospheric region of interest (represented by the yellow area in Figure 1). Five climatic fields (Relative Humidity, Specific Humidity, Air Temperature, Eastward Wind, Northward Wind) have three levels of elevation (surface, 200 and 500 hPa), totally yielding 15 climate variables. The other ten climatic fields have only one level of elevation, yielding 10 climate variables. Hence, there are 25 climate variables in our study. Each climate variable has 9 feature values extracted from the nine grid points in Figure 1. Hence, there are totally 225 features in our study. Referring to Figure 1, the total number of features, n, is 225 and the total number of groups, m, is 25 as we group the features based on their type/climate variable.

4 EXPERIMENTS AND RESULTS

4.1 Experiment Setup

We followed a split-sampling approach where the dataset is divided into training period (80%, year 1971 to 2010), and testing period (20%, 2011 to 2020). To evaluate the AIG-TRANSFORMER model, we adopted four performance metrics that are often used in the downscaling literature. These metrics are: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Percent Bias (PBIAS) and the Nash-Sutcliffe Efficiency coefficient (NSE) [20]. NSE is often used in the environmental field to study the fit between observed and simulated time series data.

Table 2: Parameters and their values used by AIG-TRANSFORMER.

Parameter	Value	Description
Optimizer	Adam	Optimization function used
Loss Function	MSE	Loss function to be optimized
Epochs	500	Number of epochs
Batch Size	64	Size of each batch
Number of Transformer Blocks	4	Number of the multi-head attention blocks
Number of Filters	16	Number of the filters within 1D convolution layer (Conv1D in Figure 2)

Table 3: Results of ablation tests.

	RMSE (^{o}F)		MAE (^{o}F)		NSE		PBias (%)	
Method	Tmax	Tmin	Tmax	Tmin	Tmax	Tmin	Tmax	Tmin
TRANSFORMER	5.102	4.963	3.941	3.701	0.81	0.82	**-0.528**	2.989
AIG	5.52	4.97	4.18	3.87	0.79	0.79	-0.87	2.35
AIG-TRANSFORMER	**4.72**	**4.43**	**3.52**	**3.34**	**0.84**	**0.85**	-0.808	**2.11**

4.2 Model Development and Calibration

Multiple dynamic parameters within AIG-TRANSFORMER exist, and one would need to perform parameter tuning to achieve the best results. As such, 20% of the training dataset is kept for validation purposes. Parameters that achieve the best performance on the validation set are chosen. These parameters and the corresponding values are shown in Table 2. Note that we developed AIG-TRANSFORMER using Keras, and any Keras-related parameter not in Table 2 is set to its default value.

4.3 Ablation Studies

We performed ablation tests by considering different variants of AIG-TRANSFORMER for downscaling the weekly averages of maximum and minimum temperatures. Here, TRANSFORMER refers to the model without the AIG part. AIG refers to the model without the transformer part, where the outputs of the AIG blocks are directly fed into fully connected layers. Table 3 shows that the full model, AIG-TRANSFORMER, outperforms the other variants in terms of RMSE, MAE and NSE. These results demonstrate the importance of both the attention-based input grouping and the transformer for the temperature downscaling problem.

4.4 Model Evaluation and Comparison

After calibration and choosing the suitable parameters, we evaluated the trained AIG-TRANSFORMER model on the testing dataset. To analyze the model performance, we compare our results with those of the existing machine learning methods commonly used in statistical downscaling. These methods include Random Forests (RF), Linear Regression (LR), Support Vector Regression (SVR), Decision Trees (DT) and Long-Short Term Memory (LSTM) Networks. We also benchmarked the results of AIG-LSTM, which represents the AIG network connected to an LSTM autoencoder. Note that for the machine learning techniques RF, LR, SVR and DT we set *lag* to 1, that is we used the climate variables at time step t to predict the temperature at that same time step. Table 4 presents experimental results for predicting the weekly averages of the maximum and minimum temperatures.

It can be seen from Table 4 that AIG-TRANSFORMER outperforms the existing machine learning methods in terms of RMSE (4.4^{o}F for Tmin and 4.7^{o}F for Tmax), MAE (3.3^{o}F for Tmin and 3.5^{o}F for Tmax), NSE (0.85 for Tmin and 0.84 for Tmax). In general, all the methods had small percent bias, indicating that there is no trend to under or over-estimate the observations. Moreover, the correlation between the predicted and observed temperatures, in terms of NSE, shows that machine leaning can reproduce the observations, with different levels of performance, ranging from an NSE of 0.668 (RF on Tmin) to an NSE of 0.85 (AIG-TRANSFORMER on Tmin). These results indicate that machine learning is a suitable approach for statistical downscaling. More specifically, our AIG-TRANSFORMER achieves an improved performance, relative to the existing methods. This good performance indicates that complex non-linear relationships exist between the GCM-simulated climatic fields and the local observations, which the existing methods do not learn well. However, we note that AIG-TRANSFORMER predicts the weekly averages of minimum and maximum temperatures well in most cases (see Figure 3a and 3b), but the model falls short of accurately predicting some extreme cases. This behavior is attributable to the fact that less observations fall in the extreme cases, which created a barrier for machine learning to learn the corresponding patterns. To overcome this behavior, one could use more data to calibrate the model, as well as to add domain-specific knowledge to the model, which we intend to pursuit in future work. To further demonstrate the ability of AIG-TRANSFORMER for the statistical downscaling problem, we recalibrated and validated the model using daily minimum and maximum temperatures, instead of the weekly averages of minimum and maximum temperatures. Figures 3c and 3d show the scatter plots between daily observed and predicted temperatures. AIG-TRANSFORMER was able to reach an NSE of 0.74 and 0.71 for Tmin and Tmax respectively, and 4.4^{o}F and 5.2^{o}F in terms of MAE for downscaling daily Tmin and Tmax, respectively.

Table 4: Performance comparison of the machine learning methods studied in the paper.

	RMSE (°F)		MAE (°F)		NSE		PBias (%)	
Method	Tmax	Tmin	Tmax	Tmin	Tmax	Tmin	Tmax	Tmin
RF	6.583	6.353	5.178	4.882	0.706	0.668	-0.208	2.703
LR	6.014	5.784	4.757	4.591	0.755	0.724	-0.396	2.37
SVR	5.76	5.117	4.279	3.901	0.775	0.784	-0.481	2.436
DT	6.849	6.24	5.105	4.816	0.682	0.679	**0.094**	3.501
LSTM	5.175	4.974	4.172	3.978	0.786	0.797	-0.283	**1.426**
AIG-LSTM	4.925	4.654	4.012	3.75	0.793	0.814	0.999	1.814
AIG-TRANSFORMER	**4.72**	**4.43**	**3.52**	**3.34**	**0.84**	**0.85**	-0.808	2.11

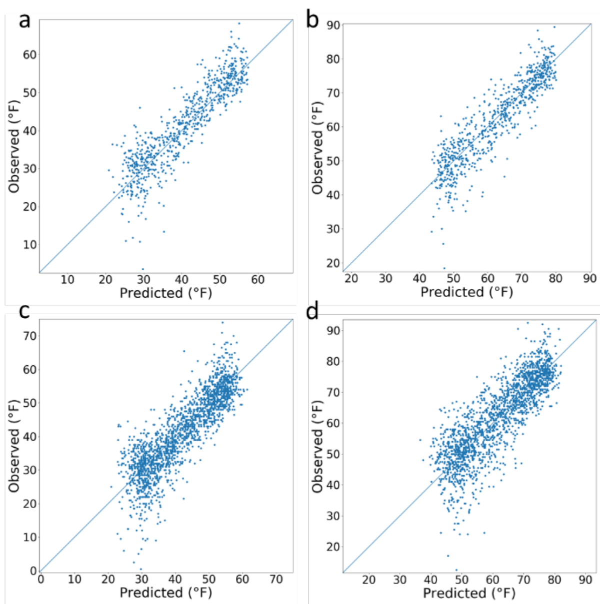

Figure 3: Scatter plots between the observed and predicted (a) weekly averages of minimum temperatures, (b) weekly averages of maximum temperatures, (c) daily minimum temperatures, and (d) daily maximum temperatures obtained by AIG-TRANSFORMER.

4.5 Long-Term Projections

The predictive performance of our proposed deep learning approach shows the suitability of such a model for downscaling both the minimum and maximum temperatures. To produce the long-term projections, we extracted future GCM simulations under the RCP8.5 scenario between 2030 and 2049 inclusively. RCP8.5 represents a high-end baseline emission scenario, depicting 8.5 W/m^2 radiative forcing in the year 2100. We used these future simulations as input to the trained AIG-TRANSFORMER model to predict the minimum and maximum temperatures between 2030 and 2049. Figure 4 shows the projected annual means of minimum and maximum temperatures. An increasing trend is shown for both cases, with the annual average reaching 63.3°F for the maximum temperature, and 41.4°F for the minimum temperature. This trend is captured using the best-fit straight line which is calculated using the least square method and provides an adequate approximation of the data.

5 CONCLUSIONS

The outputs of General Circulation Models (GCMs) are too coarse to monitor climate change at a local scale in a local region, and therefore, downscaling is required. This work proposes a deep learning approach, named AIG-TRANSFORMER, for the statistical downscaling of the weekly averages of maximum and minimum temperatures. The proposed approach consists of an attention-based input grouping (AIG) network, followed by a transformer. The AIG network employs an attention mechanism while grouping the input features of the same type (humidity, precipitation, temperature, etc.) and producing one value per group. To showcase the performance of our approach, we performed the downscaling over the Hackensack-Passaic watershed, in northeast New Jersey. Large-scale GCM-simulated climatic fields were used as the input to AIG-TRANSFORMER, which produced as output the predicted weekly averages of maximum and minimum temperatures in a local region. Experimental results demonstrated the superiority of our approach over existing machine learning methods. Moreover, our model was able to downscale the temperatures at a daily scale with good results. Finally, we used our trained AIG-TRANSFORMER to perform long-term projections for both the maximum and minimum temperatures, for the future period between 2030 and 2049. GCM simulations are available for this period and were used as input to AIG-TRANSFORMER. Efficiently downscaling temperature from the simulations of a general circulation model is crucial, as it enables us to perform long-term projections over local regions. These long-term projections enable practitioners and decision makers to better analyze and monitor climate change. In future work, we aim to compare are projections with ones produced by classical downscaling statistical approaches.

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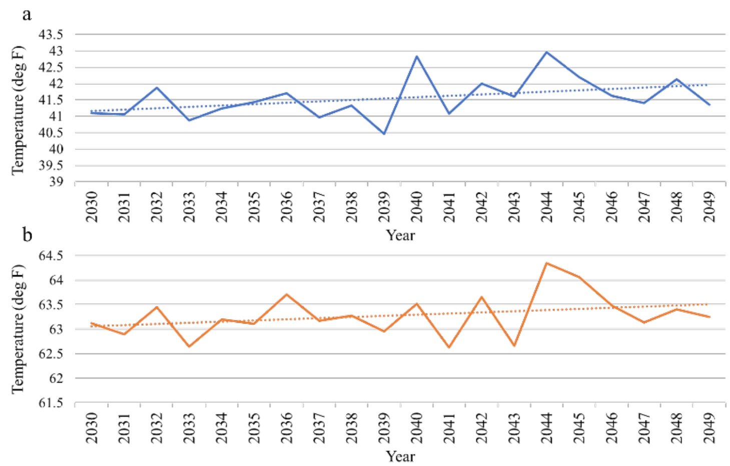

Figure 4: Projected annual means of (a) minimum and (b) maximum temperatures between 2030 and 2049 inclusively. The solid lines represent the annual means of (a) minimum and (b) maximum temperatures, and the dotted lines represent the trend lines in both charts.

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